CEBlock: Balance Factor-Driven Change Enhancement Block via Supervised Contrastive Learning for Remote Sensing Change Detection

Xinhai Li, Junyan Lu, *Member, IEEE*, Qinglei Hu, *Senior Member, IEEE*, Minghui Yang, Haobin Hua, and Kuinan Liu

***Abstract*—The scarcity of annotated datasets remains a primary obstacle for remote sensing change detection (RSCD). Most current methods address this by designing entirely new model architectures from scratch, a process that is labor-intensive and often yields only marginal performance gains. While supervised contrastive learning can enhance feature discrimination with limited labels, it suffers from two critical drawbacks in the RSCD domain: first, the computation of fine-grained pixel-wise contrastive loss on lengthy pixel sequences imposes a heavy memory burden. Second, the inherent coupling of gradient contributions from positive and negative sample pairs in changed regions weakens the discrimination of true changes. To overcome these limitations, we propose a balance factor-driven change enhancement block (CEBlock). Our solution incorporates a two-stage feature compression (TSFC) mechanism to efficiently compress lengthy bitemporal sequences, reducing memory requirements and enabling more precise identification of changes. Additionally, we introduce a piecewise loss function that uses a balance factor to replace the correlation calculation of positive and negative samples in real change regions, to separate the gradient contributions of positive and negative sample pairs in real change regions, thereby enhancing the model's attention to change features. A key advantage of the CEBlock is its plug-and-play design; it can be directly integrated into the Siamese feature extraction networks of existing models [including convolutional neural networks (CNNs), Transformers, and Mamba frameworks] to optimize feature alignment with minimal effort. Experiments show that methods augmented with our CEBlock achieve comprehensive performance enhancements on the WHU-CD, DSIFN-CD, and SYSU-CD datasets and establish new state-of-the-art (SOTA) results on two public benchmarks, with *F*1/intersection over union (IoU) improvements of 0.51%/0.77% and 3.30%/3.68% on SYSU-CD and DSIFN-CD, respectively.**

***Index Terms*—Change enhancement block (CEBlock), remote sensing change detection (RSCD), supervised contrastive learning, two-stage feature compression (TSFC).**

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I. Introduction

REMOTE sensing change detection (RSCD) is one of the important branches in the field of remote sensing. It uses sensors to capture static images of the same location at different times and employs quantitative or qualitative analysis methods to detect spatiotemporal differences in land cover, object morphology, or functions. RSCD finds applications in diverse fields, including land-use change [1], [2], [3], urban development [4], [5], and natural disaster assessment [6].

In recent years, with the significant development of deep learning technologies in basic tasks such as computer vision [7], [8] and natural language processing [9], [10], more and more people have begun to consider the application of these effective deep learning technologies in the field of remote sensing [11], [12], [13], [14], [15], [16]. In the task of bi-temporal RSCD, the mainstream methods currently adopt the Siamese network paradigm in the feature extraction stage [17], [18], [19], [20]. That is, for bi-temporal images, a feature extraction network with shared parameters is used to extract features from the bi-temporal images. Depending on the differences in task scenarios, the Siamese feature extraction network can be based on convolutional neural networks [21], [22], Transformer architectures [23], [24], or Mamba architectures [25], [26], [27]. Although these methods have achieved some promising results, they are all constrained by the limited availability of annotated remote sensing data. Moreover, to improve performance, nearly every new method requires building a change detection network from scratch, and even with substantial efforts, satisfactory results may not be obtained.

The emergence of contrastive learning offers potential solutions to mitigate this challenge. SimCLR [28] and MoCo [29] propelled contrastive learning to the forefront of self-supervised learning research through the integration of the InfoNCE loss function with large-scale data augmentation. Inspired by these advancements, an increasing number of

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researchers have introduced pixel-level supervised contrastive learning into the field of RSCD to enhance change discrimination capabilities [30].

However, the pixel sequences in multilevel images are inherently long, and directly computing contrastive learning loss on them imposes high demands on GPU memory. Moreover, most existing supervised contrastive approaches employ cross-entropy loss or its generalized triplet loss formulation. This loss computation method mathematically couples the gradient contributions from positive and negative sample pairs in the bi-temporal change regions, ultimately diminishing the model's discriminative capability for change detection.

Addressing the above issues, we first propose a balance factor-driven change enhancement block (CEBlock) with a two-stage feature compression strategy (TSFC) to reduce GPU memory requirements when computing fine-grained pixel-level contrastive learning loss by compressing the original bi-temporal image pixel sequences, as shown in Fig. 1. Second, for change detection tasks, we design a dedicated change enhanced contrast loss (CECLoss) piecewise loss function to separately handle unchanged cases and the cases of pseudo-changes amid changes (the calculated loss function is CECLoss-1) and the cases of true changes amid changes (the calculated loss function is CECLoss-2). For the loss function segment addressing the cases of true changes amid changes, we posit that if a given region changes across temporal images, its feature representations should differ significantly, with no positive samples existing. In this case, we mathematically obtain CECLoss-2 by replacing the calculation of cosine similarity between positive and negative samples in the numerator of the loss function CECLoss-1 with a balanced constant factor. Through this, we decouple the coupling relationship between positive and negative sample pairs to eliminate the contribution of false-positive samples to the gradient during backpropagation, thereby enhancing the model's attention to change features. Notably, the CEBlock directly acts on the Siamese feature extraction network of existing methods. Through contrastive learning, it enhances the similarity of feature representations in unchanged regions while increasing the discrepancy in changed regions across bi-temporal images. This, therefore, optimizes the alignment of features extracted by the Siamese network, offering a plug-and-play solution. Meanwhile, unlike methods that build change detection networks from scratch, it can improve the model's change detection capability with minimal workload. We conduct extensive experiments on three remote sensing CD datasets, demonstrating our method's superiority in enhancing the original approach's capability to extract robust features, as shown in Fig. 2. The CEBlock acts on top of the feature extraction network and before the further processing of features. Thanks to this, the CEBlock can be effectively generalized to convolutional neural networks (CNNs), Transformers, and Mamba structures. The main contributions of this article are as follows.

1) A novel TSFC effectively condenses pixel-level multiscale features from bi-temporal images, reducing memory overhead for direct feature computation and improving the discriminability between real changes and pseudo-changes in the changed regions.

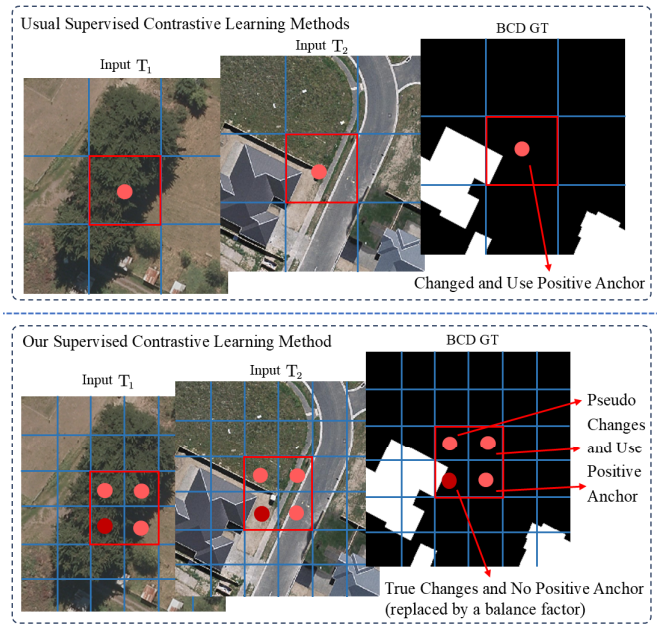


Fig. 1. Comparison between our supervised contrastive learning method and the strategies of usual supervised contrastive learning methods. Due to granularity limitations, usual supervised contrastive learning tends to identify some unchanged regions as changed ones, and these regions are regarded as pseudo-changes. The changed regions indicated by labels are recognized as real changes.

2) The proposed task-specific piecewise loss function decouples gradient contributions between positive–negative sample pairs in changed regions, significantly enhancing feature discriminability.

The rest of this article is organized as follows. Section II describes the related work. Section III gives the details of our proposed method. Some experimental results are reported in Section IV. The conclusion is made in Section V.

II. Related Work

A. BCD Models Based on Deep Learning

Due to the extensive practical demand for analyzing bi-temporal high-resolution remote sensing images and the rapid development of deep learning techniques in upstream tasks, researchers are currently focusing on deep learning-based binary change detection (BCD) techniques [21]. Based on differences in fundamental computational units, deep learning-based BCD methods can be categorized into three types: CNN-based approaches [21], [22], [38], [39], [40], [41], [42]; Transformer-based methods [23], [24], [43]; and Mamba-based frameworks [26], [27]. CNNs exhibit significant advantages in local feature extraction, while Transformers can effectively model global contextual dependencies [24], [43]. Mamba achieves global context modeling with linear computational complexity, breaking through the quadratic computation bottleneck of Transformers, thereby significantly improving processing efficiency while maintaining high accuracy. Regardless of the computational unit used, the mainstream core architecture adopts a Siamese network paradigm to handle bi-temporal change detection tasks. This Siamese structure

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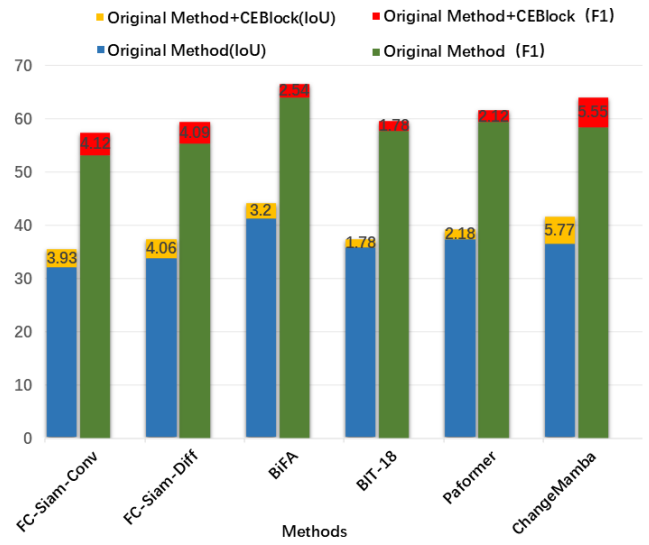


Fig. 2. Improvements in key metrics IoU and $F1$ of different methods on the DSIFN-CD dataset due to the augmentation of the CEBlock.

typically includes weight-sharing feature extraction components, and the extracted bi-temporal features form the critical foundation for subsequent change detection.

Among them, due to their weight-sharing convolution kernels, CNNs can capture the positional relationships and feature representations of image features with fewer parameters. Based on this, fully convolutional networks (FCNs) [21] were the first to be introduced into the field of change detection, including the FC-Siam-Conc model (which processes bi-temporal images through channel concatenation) and its variants FC-EF and FC-Siam-Diff (Siamese CNN architectures). Subsequent studies have optimized CNN models through various approaches: enhancing feature interaction [44], multilevel supervision [39], [40], difference representation learning [45], object-level optimization [46], [47], multiscale fusion [48], [49], and noise suppression [50], [51]. Following the successful application of Transformers in computer vision [7], [52], their powerful capability in modeling long-range dependencies was quickly adapted to change detection tasks. Chen et al. [53] proposed the BIT (bi-temporal image Transformer), the first Transformer model specifically designed for change detection, which achieves efficient contextual modeling through feature tokenization. Subsequent research has mainly expanded along three directions: multiscale feature refinement [23], [54], spatiotemporal relationship modeling [41], and global interaction enhancement [55], [56]. However, the limited receptive field of CNNs restricts their ability to model long-range dependencies, especially in scenarios with sparse change targets. The quadratic computational complexity of Transformers incurs substantial overhead, severely limiting their practicality in high-resolution remote sensing images. Emerging state-space models (e.g., Mamba) show great potential in long-sequence modeling, with linear computational complexity providing an efficient solution to the problem of long-range dependencies in visual tasks. Vision Mamba [57] addresses the issue of positional sensitivity through encoding and bidirectional scanning, while VMamba adopts a cross-scanning mechanism to enhance global context capture. Since then, Mamba architectures have rapidly expanded into fields such as medical imaging [58], [59], innovations in scanning mechanisms [60], [61], and remote sensing applications [25], [26].

All these deep learning algorithms have achieved promising results after incorporating the Siamese feature extraction network. Among them, several methods [39], [40] have also recognized the importance of feature alignment between Siamese networks for change detection tasks. However, they do not consider leveraging the inherent property of classic contrastive learning methods that are naturally suited for multibranch structural comparison to achieve supervised alignment of the Siamese feature extraction network. This constitutes the starting point for our design of the multiscale feature alignment CEBlock based on contrastive learning.

B. Change Detection Based on Contrastive Learning

Unlike traditional visual segmentation tasks, the field of RSCD requires acquiring bi-temporal annotated data simultaneously. This renders the label data used in traditional visual segmentation tasks inapplicable for change detection, constituting a core constraint on the development of the change detection field [62], [63]. Contrastive learning [28], [29] as an efficient training paradigm utilizing unlabeled or sparsely labeled data has become a crucial means to address this bottleneck in the RSCD domain.

Currently, the contrastive learning methods applied in the field of RSCD basically follow the paradigms used in upstream tasks, that is, guiding the model to learn feature extraction by controlling the relationship between positive/negative samples and anchor points in the feature projection space: pulling anchor samples closer to positive samples while pushing them away from negative sample sets [64], [65]. As can be clearly seen in Table I, these methods can be divided into supervised contrastive learning and self-supervised contrastive learning based on whether manual labels are used in the contrastive learning process. Taking CLIP [34], a landmark method for self-supervised contrastive learning applications, as the baseline, this upstream approach establishes the basic granularity scale for contrastive learning loss calculation and the coupling relationship between positive and negative samples. Building on this paradigm, SSLChange [31] explores a novel self-supervised contrastive framework, which promotes the model to accurately capture spatial, structural, and semantic information through domain adapters and hierarchical contrastive heads. DAEDFN [32] proposes a self-supervised learning method that combines denoising autoencoders with contrastive learning, providing effective data representations for change detection tasks. SSCMNet [33] designs a multiscale self-supervised constraint module to narrow the gap between self-supervised and supervised contrastive learning. While these self-supervised learning methods can extract meaningful data representations without explicit training labels, they rely on the geospatial invariance hypothesis that unchanged regions at the same location form positive sample pairs [66] and samples from different images serve as negative samples,

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TABLE I
COMPARISON OF STRATEGY DIFFERENCES AMONG SEVERAL REPRESENTATIVE CONTRASTIVE LEARNING METHODS

Type	Method	Granularity Level[1]	Multi Scale[2]	Relationship Decoupling[3]
$\mathcal{F}$[4]	SSLChange [31]	-	✘	✘
	DAEDFN [32]	-	✘	✘
	SSCMNet [33]	-	✔	✘
	CLIP [34]	32	✘	✘
$\mathcal{S}$[5]	SCPFCD [35]	-	✘	✘
	SCC [36]	32	✘	✘
	SiamContrast [37]	100(max)	✘	✘
	Ours	1024(max)	✔	✔

[1]**Granularity Level** refers to the number of image embeddings used for calculating the contrastive learning loss.
[2]**Multi Scale** refers to the utilization of multi-scale feature alignment for change detection.
[3]**Relationship Decoupling** refers to decoupling the relationships between positive and negative sample pairs in true changes.
[4]$\mathcal{F}$ denotes self-supervised contrastive learning.
[5]$\mathcal{S}$ denotes supervised contrastive learning.

which may lead the model to learn unstable feature representations. Supervised contrastive learning uses semantic labels to precisely define positive and negative samples [35], [36], [37], which can effectively address this issue. SCPFCD [35] proposes a two-stage cascaded pretraining fine-tuning paradigm based on supervised contrastive learning to initialize the Siamese network, thereby improving the network's feature extraction capability. SCC [36] leverages supervised contrastive learning to simultaneously utilize global and local information for pseudo-change augmentation. SiamContrast [37] employs supervised contrastive learning to consider both single-temporal contrastive loss and cross-historical contrastive loss, learning temporally invariant land-cover features to enhance the discriminability of time-invariant features in bi-temporal images. These methods all leverage multiscale information to enhance the expressive capability of features, employ the InfoNCE loss function and its variants to train change detection models, and have achieved satisfactory progress.

Nevertheless, as shown in Table I, when these methods apply the InfoNCE loss function to RSCD, they all strictly adhere to the loss calculation paradigm of positive–negative sample coupling. For change detection tasks, this fails to highlight the role of "making the model pay more attention to image changes" in improving the overall change detection capability of the model. Meanwhile, although these methods basically take into account the superiority of multiscale features in feature representation, they fail to leverage bi-temporal Siamese multiscale feature alignment to enhance the anti-interference capability of the Siamese feature network. Although SSCMNet [33] takes this into account, it opts to fuse the features of the bi-temporal Siamese network, which introduces interfering information for the feature alignment of the Siamese network. To address these issues, we uniformly treat bi-temporal multiscale embeddings containing changed regions as negative samples. Through TSFC (a module supporting multiscale fine-grained pixel-wise calculations) and the CECLoss function, the CEBlock mathematically separates the mutual influence of positive and negative samples in multiscale embeddings of true change regions, thereby achieving change enhancement for multiscale embeddings in true change regions and alignment of bi-temporal multiscale embeddings.

III. METHODOLOGY

A. Overview

The overall framework of our CEBlock is illustrated in Fig. 3. It comprises a feature compression module and a contrastive loss computation module. In the field of change detection, which inherently examines state variations at specific spatial locations across temporal instances—existing approaches typically extract features separately from dual-temporal imagery as their foundational step.

Our method first processes corresponding multilevel feature maps (originally extracted by the feature extraction stage) from dual-temporal inputs through feature compression. This prepares the multilevel representations for subsequent contrastive learning. The compressed features are then fed into the CECLoss module to compute multilevel contrastive learning losses. Notably, this entire process requires *no structural modifications* to the original methods, ensuring convenient and easy integration.

B. Two-Stage Feature Compression Mechanism

Considering the substantial hardware overhead associated with directly computing contrastive loss on large-scale feature maps extracted by existing models, and recognizing that projection operations before contrastive learning can enhance its effectiveness, feature compression of input feature maps becomes essential, as shown in Fig. 4.

First stage, we first apply shared tokenization to multiscale features $\{\mathbf{F}_1, \mathbf{F}_2\}$ (where $\mathbf{F} \in \mathbb{R}^{B\times H\times W\times C}$) from dual-temporal images. This process converts local features into higher-level token sequences $\mathbf{F}_{1k}$, $\mathbf{F}_{2k}$, effectively reducing computational complexity

$$\mathbf{F}_{1k} = \text{ToTokens}(\mathbf{F}_1) \tag{1}$$
$$\mathbf{F}_{2k} = \text{ToTokens}(\mathbf{F}_2). \tag{2}$$

Second stage, to enhance robustness against local variations, we apply random masking to $\mathbf{F}_{1k}$ and $\mathbf{F}_{2k}$ as a data augmentation strategy. This technique prevents model overfitting while improving generalization capability by forcing the model to focus on the most discriminative regions while maintaining global consistency

$$\mathbf{F}_{1d} = \text{PatchDropOut}(\mathbf{F}_{1k}) \tag{3}$$
$$\mathbf{F}_{2d} = \text{PatchDropOut}(\mathbf{F}_{2k}) \tag{4}$$

where $\mathbf{F}_{1d}, \mathbf{F}_{2d} \in \mathbb{R}^{B\times N\times C}$ with $N = (H/2)\times(W/2)$.

Subsequently, the masked features pass through a weight-shared 3-layer MLP with batch normalization, projecting tokens into a normalized embedding space

$$\mathbf{F}_{1p} = \text{Projector}(\mathbf{F}_{1d}) \tag{5}$$
$$\mathbf{F}_{2p} = \text{Projector}(\mathbf{F}_{2d}). \tag{6}$$

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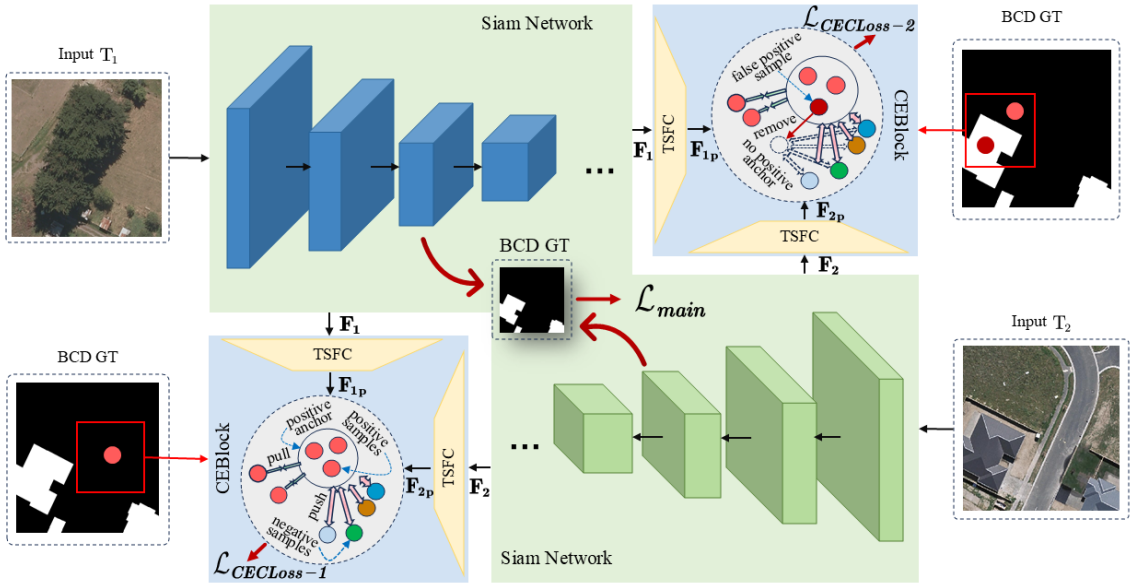


Fig. 3. Illustration of the proposed CEBlock. The dual-temporal images T_1 and T_2 are input into the Siamese network in the original method to obtain the original loss $\mathcal{L}_{\text{main}}$. The multilevel features of the dual-temporal images extracted by the original Siamese network are input into the TSFC to obtain the compressed features F_1p and F_2p. The corresponding CECLoss is calculated in two cases according to whether changes occur as specified by the labels. In the unchanged cases and the cases of pseudo-changes amid changes, the pixels in the unchanged regions of the dual-temporal images are directly identified as positive samples for each other, and the anchor in the contrastive learning loss calculation is regarded as a positive anchor, which is used to calculate $\mathcal{L}_{\text{CECLoss-1}}$, while the pixels in the cases of true changes amid changes of the dual-temporal images are identified as false-positive samples for each other, and the anchor in the contrastive learning loss calculation is regarded as a no positive anchor, which is used to calculate $\mathcal{L}_{\text{CECLoss-2}}$.

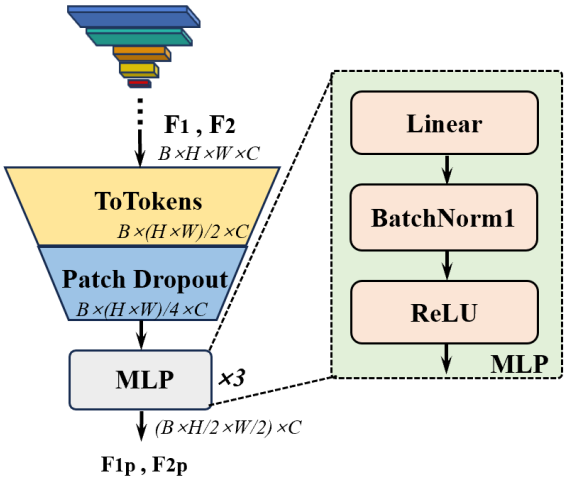


Fig. 4. Illustration of the TSFC.

During the projection process, the linear layer maps multiscale feature maps from different dimensions to a specified dimension, which further balances the computational load and performance. The BatchNorm and ReLU operations ensure feature distribution stability while preventing gradient explosion/vanishing. This projection compresses high-dimensional features to a lower-dimensional space for subsequent contrastive loss computation, resulting in $\mathbf{F}_{1p}, \mathbf{F}_{2p} \in \mathbb{R}^{B\times N\times D}$, where D denotes the projection dimension.

C. Change Enhanced Contrast Loss

The CECLoss mathematically segregates false-positive samples from positive samples within the population, treating them as negative samples during computation. The procedure initiates with feature concatenation of $\mathbf{F}_{1p}$ and $\mathbf{F}_{2p}$, followed by L2 normalization to prevent dimensional dominance in similarity measurement

$$\mathbf{L} = \text{Concat}\left(\mathbf{F}_{1p}, \mathbf{F}_{2p}\right) \tag{7}$$

$$\mathbf{L}^* = \mathcal{L}2\left(\mathbf{L}\right). \tag{8}$$

The cosine similarity matrix $\mathbf{C}$ is then computed to mitigate feature magnitude influences

$$\mathbf{L}_c = \mathrm{C}\left(\mathbf{L}^*, \mathbf{L}^*\right). \tag{9}$$

Subsequently, the ground-truth (GT) mask undergoes binarization and resizing to match the spatial dimensions of $\mathbf{F}_{1p}$ and $\mathbf{F}_{2p}$

$$\mathbf{T} = f_b\left(\mathbf{GT}\right) \tag{10}$$

where $\mathbf{T}_i \in \{0, 1\}$, with 0 denoting unchanged regions and 1 indicating changed regions; f_b represents the resizing operation based on binarization, bilinear interpolation, and patchification, which is used to adapt the GT mask to the dimensions of $\mathbf{F}_{1p}$ and $\mathbf{F}_{2p}$.

To distinguish it from the label change calibration after data loading, a transformed label map is generated via

$$\mathbf{T}^* = -\mathbf{T}. \tag{11}$$

The custom contrastive loss L_{CECLoss} is computed under $\mathbf{T}^*$ guidance through two distinct operational modes.

Case 1 ($\mathbf{T}^* \neq -1$): Corresponding to unchanged land-cover where false positives are absent, the standard contrastive loss formulation applies

$$\mathcal{L}_{\text{CECLoss}-1} = -\log \frac{\exp\left(\text{sim}\left(\mathbf{z}_i, \mathbf{z}_j\right)/\tau\right)}{\sum_{k=1}^{2N} \mathbb{1}_{[k\neq i]} \exp\left(\text{sim}\left(\mathbf{z}_i, \mathbf{z}_k\right)/\tau\right)}. \tag{12}$$

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Case 2 ($\mathbf{T}^* = -1$): For changed regions containing false positives, the numerator is replaced with a constant $\alpha = 0.1$ to mathematically isolate pseudo-positives

$$\mathcal{L}_{\text{CECLoss}-2} = -\log \frac{\alpha}{\sum_{k=1}^{2N} \mathbb{1} \exp\left(\text{sim}\left(\mathbf{z}_i, \mathbf{z}_k\right)/\tau\right)} \tag{13}$$

where $\mathbf{z}$ denotes feature embeddings, τ is the temperature parameter, N is the token count, $\mathbb{1}$ is the indicator function, and sim$(\cdot)$ is the cosine similarity.

From a mathematical perspective, this formulation ensures that the true changes in the changed regions only affect the denominator as negative samples, thereby maintaining gradient stability. Meanwhile, it makes $(\partial \mathcal{L}_{\text{CECLoss}-2})/(\partial \text{sim})$ depend solely on the changes in the denominator, and the magnitude of the calculation result of this formula can be altered by adjusting the size of the balance factor. In the context of RSCD, this corresponds to land-cover changes between temporal instances t_1 and t_2, where true changes exist in the contrastive computation and make the model pay more attention to the regions with these true changes, thus achieving change enhancement. Moreover, these regions are systematically distinguished from the changed regions considered by traditional contrastive learning methods.

Subsequently, the bidirectional loss is computed by simultaneously considering both $\mathcal{L}_{i,j}$ and $\mathcal{L}_{j,i}$, yielding the change enhanced contrast loss $\mathcal{L}_{\text{CECLoss}}$

$$\mathcal{L}_{\text{CECLoss}} = \frac{1}{2N} \sum_{i=1}^{N} \left(\mathcal{L}_{i,i+N} + \mathcal{L}_{i+N,i}\right) \tag{14}$$

where $N = \text{batch_size} \times \text{patch_count}$ denotes the total token count per batch.

Finally, $\mathcal{L}_{\text{CECLoss}}$ is combined with the primary task loss $\mathcal{L}_{\text{main}}$ through a weighted summation to obtain the final optimization objective

$$\mathcal{L} = \mathcal{L}_{\text{main}} + \lambda \mathcal{L}_{\text{CECLoss}} \tag{15}$$

where λ serves as the weighting coefficient balancing the contribution of the contrastive loss component.

IV. Experiments

A. Data Description

WHU-CD: The WHUCD dataset contains two aerial images from New Zealand's land, with a resolution of $32\,507 \times 15\,354$ pixels and a spatial resolution of 0.3 m/pixel, including approximately 22 000 independent buildings. To ensure the objectivity of comparative experiments, both the reproduced other advanced methods and the method with the augmented CEBlock adopt the same data segmentation specifications. Following the works in [73] and [74], we divided the original images into nonoverlapping chips of 256×256 pixels and divided them into 6096/762/762 for training/validation/testing.

DSIFN-CD: The deeply supervised image fusion network (DSIFN-CD) is a public BCD dataset. This dataset consists of multisource bi-temporal images with a resolution of 2 m/pixel from Google Earth, covering different cities in China, and includes changes in various land-cover objects such as buildings, roads, and farmland. To ensure fair comparison and result reproducibility, we adopted a consistent data partitioning scheme and reproduced several existing algorithms for comparison with our proposed CEBlock. These image pairs were sliced into smaller image patches of size 256×256 and divided into 14 400/1360/192 for training/validation/testing.

SYSU-CD: The SYSU-CD dataset contains 20 000 pairs of high-resolution aerial images with a size of 256×256 pixels, which were captured in Hong Kong from 2007 to 2014. This dataset includes various types of changes, such as new urban building construction, suburban expansion, preconstruction infrastructure work, vegetation changes, road expansion, and offshore construction. The SYSU-CD dataset basically covers most types of changes occurring in urban environments and is of great significance for detecting urban changes. To ensure fair comparison and result reproducibility, we adopted a consistent data partitioning scheme and reproduced several existing algorithms for comparison with our proposed CEBlock. To ensure fair comparison and result reproducibility, we adopted a consistent data partitioning scheme and reproduced several existing algorithms for comparison with our proposed CEBlock. These image pairs were sliced into smaller image patches of size 256×256 and divided them into 12 000/4000/4000 for training/validation/testing.

B. Implementation Details

1) Architecture Details: In the TSFC stage, the dropout random masking rate is set to 0.5. The input channel count of the projector is determined by the original channel dimensions of each hierarchical level in the bi-temporal features, while its output channel count is fixed at 256. The settings of the loss weight factors λ_1 and λ_2 are completely retained from the original method. The number of multilayer perceptrons is set to 3. In the CEBlock stage, the number of hierarchical levels participating in the computation of the custom contrastive learning loss can be dynamically adjusted according to practical requirements. Within the custom contrastive learning loss function, the numerator parameter α can be dynamically configured based on the evolving embedding space, specific methodology, and dataset characteristics; we set this value to 0.1. The magnitude of the custom contrastive learning loss should, in principle, not exceed the total loss of the original model. We maintain an approximate ratio of 1:1 between them.

2) Training Details: The proposed CEBlock is implemented based on the PyTorch framework and runs on an NVIDIA RTX 4090. Optimizer and learning rate configurations remain unchanged from those of the original model. For each comparative experiment across all datasets, the batch size is set to 8. Training epochs are set to 200 for all comparative methods, except for the AFCF3D-Net [67] method, which uses 100 epochs. The loss function for both the original model and the comparative experiments is the addition of the cross-entropy loss and the dice loss [75]

$$\mathcal{L}_{\text{main}} = \lambda_1 \mathcal{L}_{\text{ce}} + \lambda_2 \mathcal{L}_{\text{dice}} \tag{16}$$

where λ_1 and λ_2 denote the coefficients of the loss function.

3) Evaluation Metrics: For the performance measurement of similarity between the predictions and the GT, we introduce

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TABLE II

QUANTITATIVE RESULTS ON THE WHU-CD DATASET. IN THIS AND OTHER TABLES, $\mathcal{C}$ INDICATES CNN-BASED MODELS, $\mathcal{T}$ INDICATES TRANSFORMER-BASED MODELS, AND $\mathcal{M}$ INDICATES MAMBA-BASED MODELS. THE HIGHER RESULT IS HIGHLIGHTED IN BOLD

Type	Method	OA	mF1	mIoU	IoU	F1	Rec.	Pre.	Kappa
$\mathcal{C}$	FC-Siam-Diff [21]	98.96	93.12	87.78	76.64	86.77	85.36	88.24	86.23
	FC-Siam-Diff+CEBlock	**99.05**	**93.71**	**88.72**	**78.44**	**87.92**	**86.77**	**89.12**	**87.42**
	FC-Siam-Conc [21]	98.84	92.59	86.96	75.11	85.79	87.23	84.39	85.18
	FC-Siam-Conc+CEBlock	**98.84**	**92.60**	**86.96**	**75.13**	**85.80**	**87.68**	**84.40**	**85.20**
	AFCF3D-Net [67]	99.17	94.42	89.88	80.62	89.27	86.83	91.85	88.84
	AFCF3D-Net+CEBlock	**99.25**	**94.94**	**90.75**	**82.28**	**90.28**	**86.92**	**93.90**	**89.89**
	SNUNet [44]	98.56	90.86	84.34	70.17	82.47	**84.53**	80.52	81.72
	SNUNet+CEBlock	**98.63**	**91.13**	**84.74**	**70.90**	**82.97**	83.57	**82.38**	**82.26**
	DARNet [68]	98.72	91.99	86.03	73.39	84.66	88.28	81.32	83.99
	DARNet+CEBlock	**98.89**	**92.92**	**87.46**	**76.08**	**86.42**	**93.99**	**91.90**	**85.84**
$\mathcal{T}$	BIT-18 [53]	98.67	91.27	84.96	71.29	83.24	82.68	83.81	82.54
	BIT-18+CEBlock	**98.83**	**92.45**	**86.74**	**74.68**	**85.50**	**86.03**	**86.03**	**84.90**
	BIT-34 [53]	98.55	91.05	84.61	70.73	82.85	87.64	78.56	82.10
	BIT-34+CEBlock	**98.73**	**92.09**	**86.18**	**73.68**	**84.85**	**88.80**	**81.23**	**84.19**
	BIT-50 [53]	98.08	91.60	85.44	72.24	83.89	85.97	81.90	83.20
	BIT-50+CEBlock	**98.34**	**92.62**	**87.00**	**75.21**	**85.85**	**88.36**	**83.49**	**85.25**
	Bifa [55]	99.34	95.62	91.90	84.48	91.59	**89.94**	93.30	91.25
	Bifa+CEBlock	**99.36**	**95.74**	**92.10**	**84.86**	**91.81**	89.52	**94.22**	**91.47**
$\mathcal{M}$	ChangeMamba-Tiny [26]	99.21	90.00	90.50	81.82	90.00	**88.34**	91.73	89.59
	ChangeMamba-Tiny+CEBlock	**99.23**	**90.19**	**90.67**	**82.13**	**90.19**	88.01	**92.48**	**89.79**
	ChangeMamba-Small [26]	99.08	88.39	89.13	79.20	88.39	87.23	**89.58**	87.92
	ChangeMamba-Small+CEBlock	**99.13**	**89.10**	**89.72**	**80.34**	**89.10**	**89.13**	89.07	**88.64**
	ChangeMamba-Base [26]	99.07	88.18	88.95	78.85	88.18	86.82	89.58	87.69
	ChangeMamba-Base+CEBlock	**99.18**	**89.49**	**90.07**	**80.98**	**89.49**	**87.71**	**91.35**	**89.07**

seven metrics, including precision (Pre), recall (Rec), $F1$-score ($F1$), intersection over union (IoU), overall accuracy (OA), false alarm (FA), and Kappa. The metrics can be individually defined as follows:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{17}$$

$$\text{Kappa} = \frac{\text{OA} - P}{1 - P} \tag{18}$$

where

$$P = \frac{(\text{TP} + \text{FN})(\text{TP} + \text{FP}) + (\text{TN} + \text{FN})(\text{TP} + \text{TN})}{(\text{TP} + \text{FP} + \text{TN} + \text{FN})^2}. \tag{19}$$

TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively. P in Kappa denotes the hypothetical probability of the chance agreement between reference and prediction. It is worth noting that $F1$ and IoU can better reflect the generalization ability of the model.

C. Performance Comparison

To verify the validity of the CEBlock in the remote sensing CD task, in this section, several state-of-the-art (SOTA) models are chosen as the competitors, including seven methods based on convolution (FC-Siam-Diff [21], FC-Siam-Conc [21], SNUNet [44], Paformer [69], DARNet [68], AFCF3D-Net [67], and Changer [71]), two methods based on the Transformer (BIT [53] and Bifa [55]), and four methods based on Mamba (ChangeMamba [26], CDMamba [27], M-CD [72], and 3D-SSM [70]).

1) Quantitative Results: Numerically, Tables II–IV show the overall performance of all original methods and the methods with the augmentation of the CEBlock on the WHU-CD, DSIFN-CD, and SYSU-CD test sets. Bold numbers represent the best results. Evidently, regardless of whether the framework is based on the CNN, Transformer, or Mamba, the methods with the augmentation of the CEBlock exhibit superior performance compared to other original methods. Although in some methods across the three datasets, the original methods outperform those with the CEBlock in terms of the Rec. or Pre. metric, the methods with the CEBlock achieve greater improvements in the corresponding Pre., Rec., and $F1$ metrics. Notably, on the DSIFN-CD dataset, the IoU/$F1$ values of ChangeMamba-Small with the augmentation of the CEBlock are 5.77%/5.55% higher than those of the original ChangeMamba-Small. In addition, as can be seen from Table V, the key indicators IoU/$F1$ values of the methods with the augmentation of the CEBlock exceed those of the latest 3D-SSM by different percentages across various datasets. Specifically, on the SYSU-CD and DSIFN-CD datasets, the IoU/$F1$ values of the methods with the CEBlock are 0.93%/0.62% and 2.7%/2.58% higher than those of 3D-SSM, respectively. The results indicate that although the Rec. metric of 3D-SSM+CEBlock on DSIFN-CD is lower than that of 3D-SSM, the former outperforms the latter in other metrics, suggesting that it has higher accuracy in detecting changes

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TABLE III
QUANTITATIVE RESULTS ON THE DSIFN-CD DATASET. THE HIGHER RESULT IS HIGHLIGHTED IN BOLD

Type	Method	OA	mF1	mIoU	IoU	F1	Rec.	Pre.	Kappa
$\mathcal{C}$	FC-Siam-Diff [21]	81.64	71.86	58.74	38.18	55.27	72.68	44.59	44.54
	FC-Siam-Diff+CEBlock	**83.91**	**74.67**	**61.99**	**42.21**	**59.36**	**75.28**	**49.00**	**49.89**
	FC-Siam-Conc [21]	76.76	68.86	54.72	36.21	53.17	**84.53**	38.78	40.42
	FC-Siam-Conc+CEBlock	**81.38**	**72.69**	**59.43**	**40.14**	**57.29**	80.03	**44.61**	**46.58**
	Paformer [69]	83.37	74.45	61.63	42.20	59.35	**77.79**	47.80	49.63
	Paformer+CEBlock	**85.83**	**76.40**	**64.20**	**44.38**	**61.47**	72.44	**53.39**	**53.03**
$\mathcal{T}$	BIT-18 [53]	86.70	74.90	62.96	40.54	57.69	58.09	57.31	49.81
	BIT-18+CEBlock	**87.27**	**75.96**	**64.14**	**42.32**	**59.47**	**59.06**	**59.10**	**51.92**
	BIT-34 [53]	86.83	75.66	63.73	42.03	59.18	**61.20**	57.30	51.34
	BIT-34+CEBlock	**88.83**	**77.03**	**65.60**	**43.43**	**60.56**	54.96	**67.43**	**54.13**
	Bifa [55]	86.55	77.84	65.86	46.70	63.94	76.41	54.97	55.95
	Bifa+CEBlock	**87.44**	**79.38**	**67.77**	**49.80**	**66.48**	**79.80**	**56.98**	**59.02**
$\mathcal{M}$	ChangeMamba-Tiny [26]	88.10	60.22	65.00	43.08	60.22	**57.72**	62.94	53.23
	ChangeMamba-Tiny+CEBlock	**89.73**	**63.51**	**67.63**	**46.54**	**63.52**	57.29	**71.25**	**57.26**
	ChangeMamba-Small [26]	87.30	58.38	63.64	41.22	58.38	57.08	59.74	50.89
	ChangeMamba-Small+CEBlock	**89.50**	**63.93**	**67.70**	**46.99**	**63.93**	**59.65**	**68.88**	**57.82**
	ChangeMamba-Base [26]	88.66	62.78	66.60	45.75	62.78	**61.29**	64.34	56.09
	ChangeMamba-Base+CEBlock	**89.46**	**63.71**	**67.57**	**46.75**	**63.71**	59.26	**68.88**	**57.59**
	3D-SSM [70]	86.09	76.07	63.94	43.45	60.58	68.48	54.31	52.27
	3D-SSM+CEBlock	**87.38**	**77.77**	**66.00**	**46.15**	**63.16**	**69.30**	**58.01**	**55.62**

TABLE IV
QUANTITATIVE RESULTS ON THE SYSU-CD DATASET. THE HIGHER RESULT IS HIGHLIGHTED IN BOLD

Type	Method	OA	mF1	mIoU	IoU	F1	Rec.	Pre.	Kappa
$\mathcal{C}$	Changer [71]	91.21	87.24	78.08	66.86	80.14	75.20	85.76	81.72
	Changer+CEBlock	**91.50**	**87.74**	**78.81**	**67.99**	**80.94**	**76.54**	**85.89**	**82.26**
	Paformer [69]	90.43	86.04	76.33	64.21	78.20	72.76	**84.52**	72.12
	Paformer+CEBlock	90.29	**86.20**	**76.52**	**64.87**	**78.69**	**76.06**	81.51	**72.41**
$\mathcal{T}$	BIT-18 [53]	89.66	85.47	75.44	63.49	77.67	**76.26**	79.13	70.94
	BIT-18+CEBlock	**89.96**	**85.72**	**75.82**	**63.85**	**77.93**	75.17	**80.91**	**71.45**
	BIT-34 [53]	89.81	85.60	75.63	63.67	77.80	75.69	80.04	71.20
	BIT-34+CEBlock	**90.21**	**86.18**	**76.48**	**64.90**	**78.72**	**76.75**	**80.78**	**72.37**
	BIT-50 [53]	90.36	86.12	76.42	64.53	78.44	**74.38**	82.98	72.26
	BIT-50+CEBlock	**90.57**	**86.25**	**76.64**	**64.67**	**78.55**	73.19	**84.75**	**72.54**
$\mathcal{M}$	ChangeMamba-Tiny [26]	90.79	79.70	77.50	66.26	79.70	76.72	**82.93**	73.76
	ChangeMamba-Tiny+CEBlock	**90.98**	**80.65**	**78.23**	**67.58**	**80.65**	**79.72**	81.61	**74.77**
	ChangeMamba-Small [26]	90.56	79.76	77.37	66.34	79.76	**78.90**	80.65	73.61
	ChangeMamba-Small+CEBlock	**90.71**	**79.98**	**77.61**	**66.43**	**79.98**	78.66	**81.34**	**73.93**
	ChangeMamba-Base [26]	90.81	80.01	77.76	66.78	80.01	78.32	81.93	72.24
	ChangeMamba-Base+CEBlock	**91.38**	**81.44**	**79.03**	**68.69**	**81.44**	**80.23**	**82.69**	**73.07**
	3D-SSM [70]	92.60	89.66	81.70	72.62	84.14	**83.29**	85.01	79.31
	3D-SSM+CEBlock	**92.94**	**90.08**	**82.39**	**73.55**	**84.76**	83.13	**86.39**	**80.17**

in regions and is more comprehensive in detecting changing areas. In summary, the above quantitative analysis proves that the pixel-level segmented contrastive learning loss calculation of the CEBlock is crucial for CD tasks.

2) Qualitative Results: To further illustrate the validity of our proposed method, qualitative analyses are conducted on the WHU-CD, DSIFN-CD and SYSU-CD test sets (Figs. 5–7), where distinct colors are assigned to identify the correctness or incorrectness of the detection, including TP (white), TN (black), FP (red), and FN (green).

3) Visualization on WHU-CD (Fig. 5): Several representative samples are selected for visualization comparison, such as the surface disturbance caused by the cement ground covering around sparse residential areas in Fig. 5(a), and the dense changes in the entire area due to newly added densely distributed small buildings in Fig. 5(b). It can be clearly seen from the visualized data presented in Fig. 5(a-plus) and (b-plus) that the method with our proposed CEBlock augmented retains the original method's ability to identify changed regions in bi-temporal remote sensing images. Although DARNet with

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TABLE V

QUANTITATIVE RESULTS OF SOTA METHODS AT DIFFERENT STAGES ON THE TWO CD DATASETS. IN THE SYSU-CD DATASET, "OURS" REFERS TO 3D-SSM+CEBLOCK, WHILE IN THE DSIFN-CD DATASET, "OURS" REFERS TO AFCD3D-NET+CEBLOCK. THE HIGHER RESULT IS HIGHLIGHTED IN BOLD

Type	Method	SYSU-CD						DSIFN-CD					
		OA	Pre.	Rec.	IoU	F1	Kappa	OA	Pre.	Rec.	IoU	F1	Kappa
$\mathcal{C}$	FC-Siam-Conc [21]	90.36	81.22	76.91	65.30	79.01	72.76	76.76	38.78	84.53	36.21	53.17	40.42
	FC-Siam-Diff [21]	90.27	78.19	81.46	66.37	79.79	73.38	81.64	44.59	72.68	38.18	55.27	44.54
	Paformer [69]	90.43	84.52	72.76	64.21	78.20	72.12	83.37	47.80	**77.79**	42.20	59.35	49.63
	SNUNet [44]	89.74	86.83	66.59	60.49	75.38	69.04	86.49	55.09	72.71	45.65	62.69	54.63
$\mathcal{T}$	BIT-18 [53]	89.66	79.13	76.26	63.49	77.67	70.94	86.70	57.31	58.09	40.54	57.69	49.81
	BIT-34 [53]	89.81	80.04	75.69	63.67	77.80	71.20	86.83	57.30	61.20	42.03	59.18	51.34
	BIT-50 [53]	90.36	82.98	74.38	64.53	78.44	72.37	87.78	59.04	70.95	47.54	64.45	57.15
	BIT-101 [53]	90.73	85.89	72.63	64.89	78.71	72.54	86.75	57.19	60.01	41.41	58.56	50.68
	Bifa [55]	92.73	86.10	82.47	72.78	84.25	79.52	86.55	54.97	76.41	46.70	63.94	55.95
$\mathcal{M}$	ChangeMamba [26]	92.12	88.09	77.02	69.76	82.18	77.16	87.46	58.14	70.24	46.65	63.62	56.13
	CDMamba [27]	90.01	81.80	74.51	63.91	77.98	71.59	88.70	68.25	51.56	41.58	58.74	52.34
	M-CD [72]	92.82	**88.70**	79.69	72.35	83.96	79.35	86.03	57.20	70.73	46.25	63.25	54.75
	3D-SSM [70]	92.60	85.01	**83.29**	72.62	84.14	79.31	86.09	54.31	68.48	43.45	60.58	52.27
	Ours	**92.94**	86.39	83.19	**73.55**	**84.76**	**80.17**	**88.82**	**73.03**	63.18	**51.22**	**67.75**	**61.03**

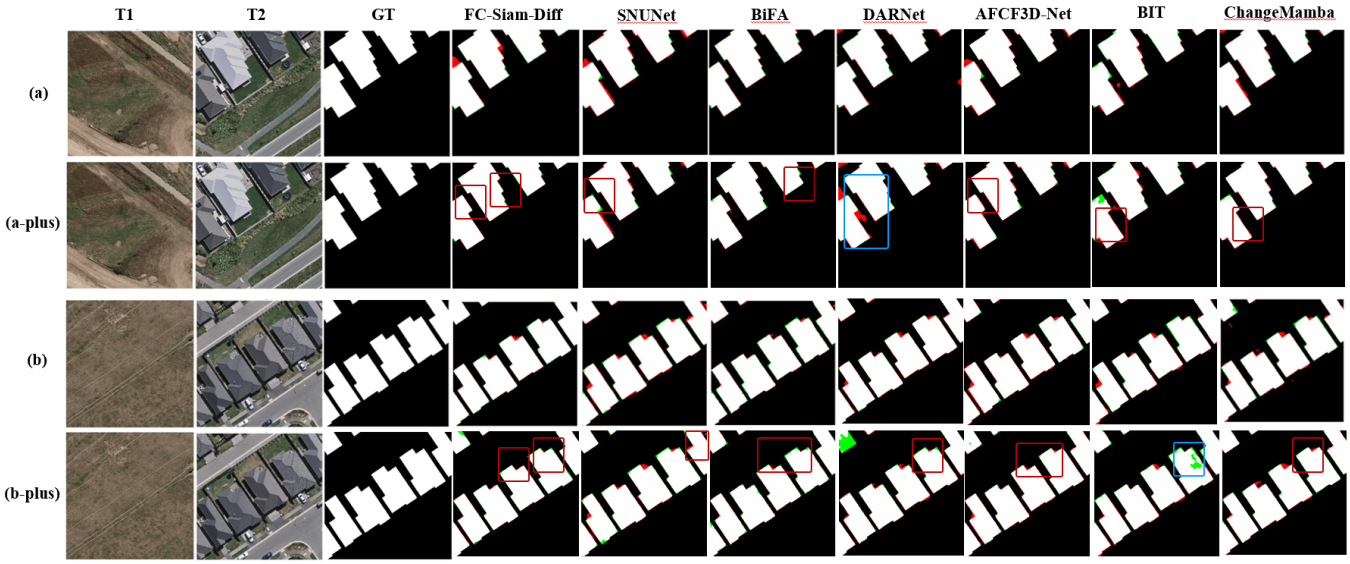


Fig. 5. Visualization results of different methods on the WHU-CD test set. (a) and (b) Representative samples. (a-plus) and (b-plus) Corresponding samples after adding the CEBlock. White represents true positives, black represents true negatives, red indicates false positives, and green stands for false negatives.

the CEBlock augmented shows a slight performance decline in some cases, its overall performance is still inferior to the method with our proposed module augmented. The results in Fig. 5(b-plus) indicate that, as can be seen from Fig. 5(a-plus), compared with its original method, the method with our proposed CEBlock augmented can improve the original method's ability to identify real changes in discontinuous but densely changing scenes. This is reflected in the fact that the method with our proposed CEBlock augmented can accurately identify narrow, unchanged gaps in densely changing regions and has clearer edges for the identified regular changing regions. This verifies the ability of the CEBlock to distinguish fine-grained pixel-level changes. In conclusion, the results of the qualitative analysis are consistent with the quantitative analysis in Table I, and the proposed CEBlock achieves the effect of enhancing the performance of the original method on the WHU-CD dataset.

4) Visualization on DSIFN-CD (Fig. 6): In DSIFN-CD, we also selected several representative samples. Fig. 6(a) shows small-scale building changes, where changes in large areas of farmland constitute the main interference. Fig. 6(b) presents a large-scale building change scenario. It can be seen that for both large-scale and small-scale changes, the method incorporating our proposed CEBlock reduces false detections compared with the original method. Notably, Fig. 6(b-plus) shows that the original method augmented by the CEBlock can improve the original method's ability to detect large-area changes, which may be due to the separation of the gradient contributions of positive and negative sample pairs in real change regions by CEBlock. Meanwhile, regarding the ability to detect narrow, unchanged gaps in large-area change scenarios, this is reflected in the fact that the original method, augmented by the CEBlock, reduces false detections of narrow,

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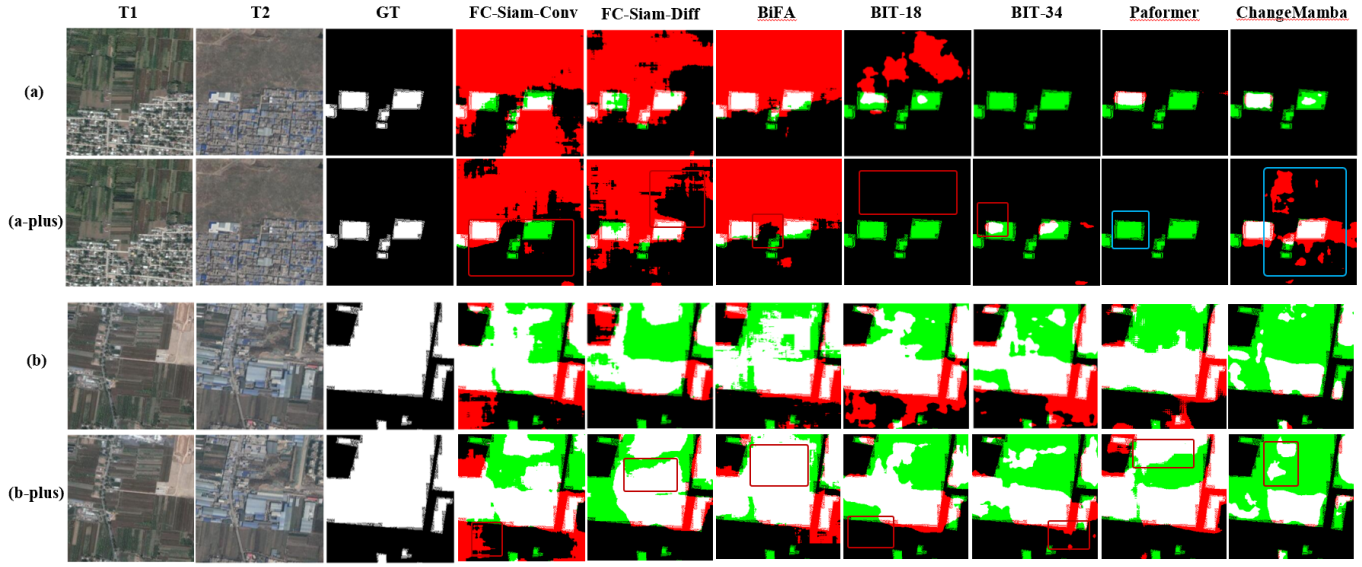


Fig. 6. Visualization results of different methods on the DSIFN-CD test set. (a) and (b) Representative samples. (a-plus) and (b-plus) Corresponding samples after adding the CEBlock. White represents true positives, black represents true negatives, red indicates false positives, and green stands for false negatives.

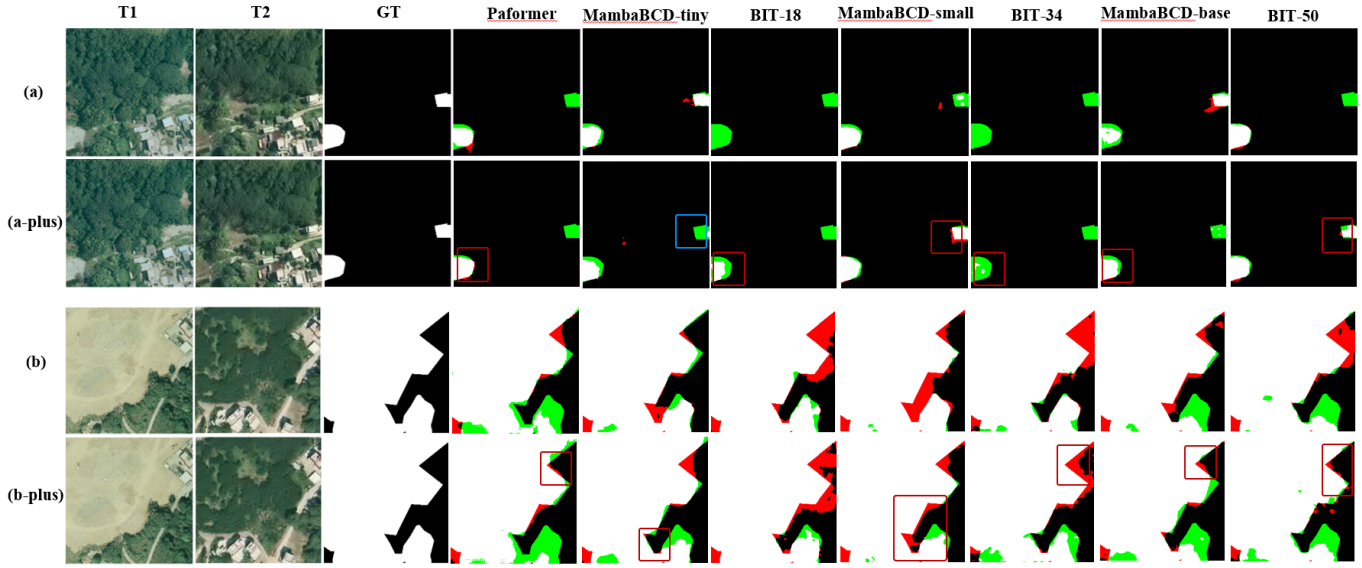


Fig. 7. Visualization results of different methods on the SYSU-CD test set. (a) and (b) Representative samples. (a-plus) and (b-plus) Corresponding samples after adding the CEBlock. White represents true positives, black represents true negatives, red indicates false positives, and green stands for false negatives.

unchanged gaps and achieves clearer detected change edges. This once again verifies the ability of the CEBlock to distinguish fine-grained pixel-level changes.

5) Visualization on SYSU-CD (Fig. 7): To further verify the detection performance for changes in various categories, we validated the CEBlock on the SYSUCD dataset, which covers a broader range of categories. It can be observed that the method with the augmentation of the CEBlock achieves better visualization results. For instance, as shown in Fig. 7(a) and (a-plus), in small-scale change detection, the BIT method integrated with the CEBlock exhibits a lower false detection rate compared to the original BIT method. Meanwhile, as illustrated in Fig. 7(b) and (b-plus), for large-scale changes caused by vegetation coverage, the change boundaries are clearer, and there are fewer false detections.

Notably, due to the inherent performance bias of the original method itself, the original method after CEBlock augmentation still fails to achieve comprehensive performance improvement in every pair of images under each scenario, as indicated by the parts marked with blue boxes in Figs. 5–7.

D. Analysis of Differences in Contrastive Learning Strategies

In Table VI, both SCPFCD and CEBlock are strategies that augment the feature recognition capability of the base method's encoder via supervised contrastive learning. SCPFCD requires a large-scale pretrained dataset to initialize the encoder of the base method; its feature contrast is confined to a single scale and fails to decouple the calculation of positive–negative sample pairs within changed regions. In contrast, the CEBlock leverages multiscale contrast and

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TABLE VI
COMPARISON OF STRATEGY DIFFERENCES AMONG REPRESENTATIVE CONTRASTIVE LEARNING METHODS ON THE WHU-CD DATASET

Type	Method	Pretraining Dataset	Contrastive Strategy	Granularity Level	Multi Scale	Relationship Decoupling	Rec	Pre	IoU	F1
$\mathcal{F}$	DAEDFN [32]	xBD	-	-	✘	✘	-	-	78.10	87.60
	SSCMNet [33]	✘	-	-	✔	✘	**92.23**	87.65	81.62	89.88
$\mathcal{S}$	FC-Siam-Conc-s[1] [21]	ImageNet	-	-	-	-	88.06	81.25	73.18	84.52
	FC-Siam-Conc-s[1] [21]	ImageNet	SCPFCD [35]	-	✘	✘	84.62	85.46	73.97	85.04
	FC-Siam-Conc-c[2] [21]	✘	**CEBlock**	1024(max)	✔	✔	87.68	84.40	75.13	85.80
	Bifa [55]	✘	**CEBlock**	1024(max)	✔	✔	89.52	**94.22**	**84.86**	**91.81**

[1] FC-Siam-Conc-s employs a 17-layer standard Conv3×3 encoder.
[2] FC-Siam-Conc-c adopts a 10-layer standard Conv3×3 encoder.

TABLE VII
TRAINING TIME (MINUTES PER 100 ITERATIONS) AND GPU OVERHEAD (GB) OF OUR CEBLOCK. ALL RESULTS ARE EVALUATED ON THE WHU-CD DATASETS

Type	TSFC	Training Time	GPU Overhead
CNN-Based	without	**0.24**	31.55
	with	0.42	**4.14**
Transformer-Based	without	42.26	**0.41**
	with	**5.96**	0.43
Mamba-Based	without	0.76	5.45
	with	**0.68**	**2.60**

decoupling strategies, which enable it to boost the key metric $F1$-score of the base method FC-Siam-Conc to 85.80 without relying on a large-scale initialized pretrained dataset. However, constrained by the inherent feature extraction capability of the base method, even with the encoder augmentation strategy based on supervised contrastive learning, there remains a certain performance gap compared with SOTA self-supervised contrastive learning methods. Nevertheless, when the base method itself possesses sufficient feature extraction capability, the base method augmented with the CEBlock can still achieve superior performance (the $F1$ of Bifa augmented by the CEBlock is 1.93% higher than that of SSCMNet). This demonstrates the superiority of the supervised contrastive learning strategy designed by the CEBlock over other contrastive learning strategies.

E. Computational Complexity

As shown in Table VII, the evaluation results of training time and GPU overhead demonstrate that our proposed TSFC module achieves an optimal balance between training efficiency and resource consumption across different architectures. Specifically, the most significant improvement is observed in the Transformer-based architecture, where the training time is drastically reduced from 42.26 to 5.96 min after integrating the CEBlock, representing an 86% efficiency improvement, while the GPU overhead increases only marginally by 4.9%. For the CNN-based architecture, although the training time increases from 0.24 to 0.42 min, the GPU memory usage is significantly reduced by 87% (from 31.55 to 4.14 GB). The Mamba-based architecture achieves dual optimization in both training time and GPU overhead, with reductions of 10.5% and 52%, respectively. These results fully demonstrate that the CEBlock, as a plug-and-play lightweight module, can effectively enhance the training efficiency of different deep learning architectures and optimize resource utilization without introducing significant computational burden, exhibiting excellent versatility and practicality.

F. Ablation Studies

To further verify the influence of each module on the performance of CD, ablation experiments are conducted in this section, as shown in Table VIII.

1) Effects of Different Components in the CEBlock: To validate the effectiveness of each component in the CEBlock, we conducted ablation experiments on the WHU dataset. All experiments were performed by incrementally adding structures to the baseline model to ensure reliability. As shown in Table VIII, through combinatorial configurations of components and parameters, the CEBlock achieves significant improvements across all three architectures. Specifically, on CNN-based architecture, the CEBlock improves $F1$ by 0.77% and IoU by 1.22% over the baseline; on Mamba-based architecture, it achieves more substantial gains of 1.91% in $F1$ and 2.66% in IoU; while on Transformer-based architecture, the improvements are 1.42% in $F1$ and 2.19% in IoU. These results demonstrate the efficacy of the CEBlock's components and parameter design, confirming that the proposed method consistently outperforms the original approaches across different architectural paradigms in change detection tasks.

2) Effects of Balance Factor α: To further validate the effectiveness of the balance factor α configuration, Table VIII demonstrates consistent trends across all three architectures. For CNN-based models, when $\alpha = 0.01$, the key performance metrics $F1$/IoU decrease by 0.18%/0.30%, while setting $\alpha = 1$ leads to more significant reductions of 7.33%/10.81% in $F1$/IoU. Similarly, for Mamba-based architecture, $\alpha = 0.01$ causes $F1$/IoU to decrease by 1.20%/1.68%, while $\alpha = 1$ results in reductions of 0.60%/0.84%. On Transformer-based models, $\alpha = 0.01$ leads to 0.30%/0.47% decreases in $F1$/IoU, while $\alpha = 1$ causes more substantial

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TABLE VIII

ABLATION STUDY OF DIFFERENT COMPONENTS OF THE CEBLOCK ACROSS THREE ARCHITECTURES. BASELINE REFERS TO THE METHOD WITHOUT ANY CONTRASTIVE LEARNING, α IS A KEY PARAMETER OF THE LOSS FUNCTION, AND LEVEL DENOTES THE MULTILEVEL FEATURES SELECTED FOR CALCULATING THE CONTRASTIVE LEARNING LOSS, WHERE LOW REPRESENTS LOW-LEVEL FEATURES AND HIGH REPRESENTS HIGH-LEVEL FEATURES

Method	α	Scale Level[1]	CNN-Based				Mamba-Based				Transformer-Based			
			IoU	F1	Rec	Pre	IoU	F1	Rec	Pre	IoU	F1	Rec	Pre
Baseline	-	-	77.15	87.10	85.66	88.59	65.46	79.13	81.82	76.61	74.49	85.38	78.60	93.45
CEBlock	0.01	4^2	78.07	87.69	86.33	**89.08**	66.44	79.84	84.38	75.76	76.21	86.50	**82.09**	91.42
	1	4^2	66.34	79.77	75.14	85.00	67.28	80.44	**86.45**	75.21	71.84	83.61	78.32	89.67
	0.1	2 (low)[3]	76.66	86.79	86.68	86.90	64.85	78.68	76.12	81.42	75.76	86.21	81.59	91.39
	0.1	2 (high)[4]	76.54	86.71	85.68	87.77	66.65	79.98	74.40	**86.48**	71.03	83.06	77.16	89.94
	0.1	**4^2**	**78.37**	**87.87**	**87.13**	88.63	**68.12**	**81.04**	80.65	81.43	**76.68**	**86.80**	80.42	**94.29**

[1] A corresponding relationship exists between Scale level and Granularity level.
[2] Scale level 4 is 64×64 + 32×32 + 16×16 + 8×8, corresponding to Granularity levels 1024, 256, 64 and 16 respectively.
[3] Scale level 2 (low) is 64×64 + 32×32, corresponding to Granularity levels 1024 and 256 respectively.
[4] Scale level 2 (high) is 16×16 + 8×8, corresponding to Granularity levels 64 and 16 respectively.

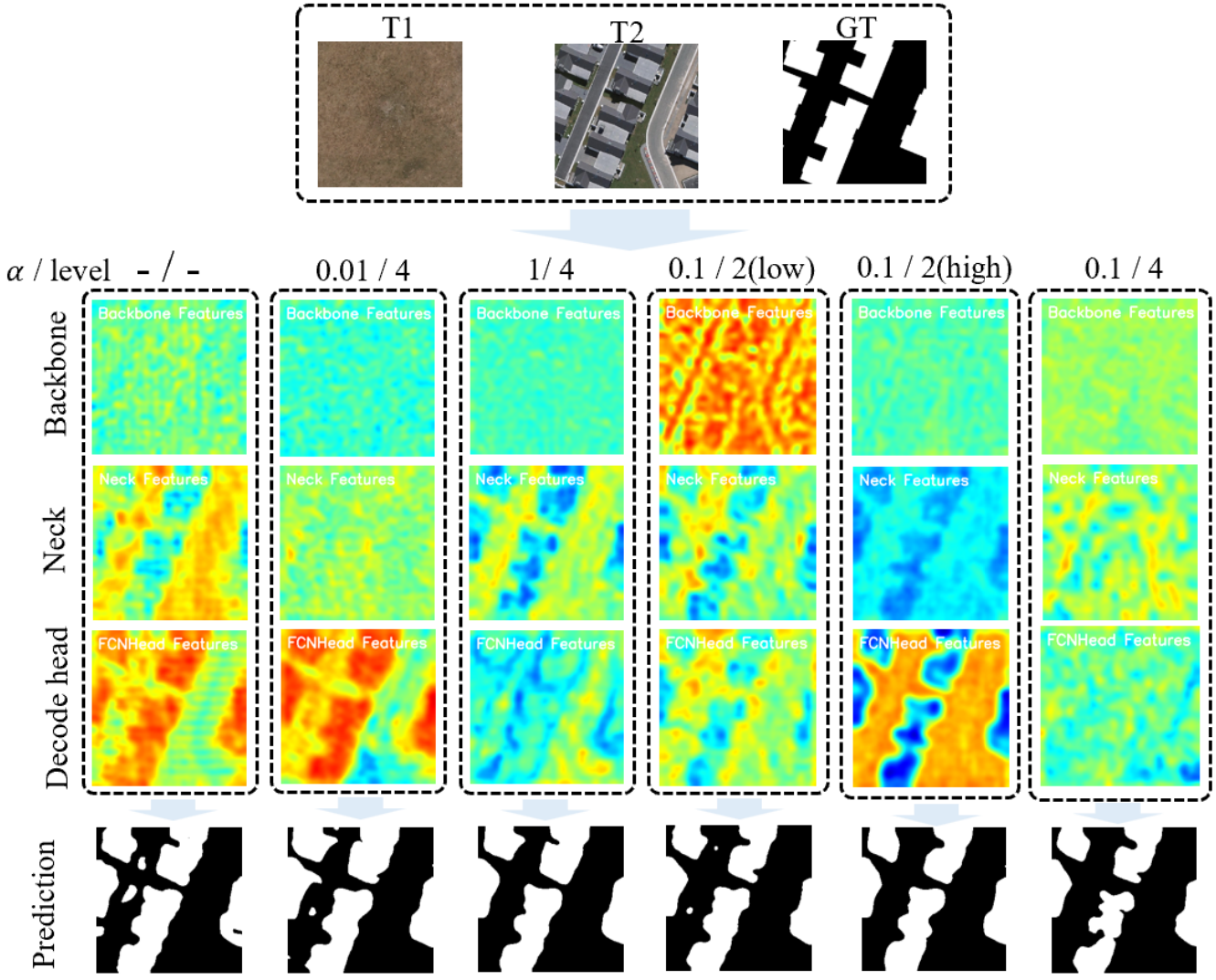


Fig. 8. Heatmap visualization results of different Transformer-based methods on the WHU-CD dataset.

reductions of 3.19%/4.83%. These results conclusively verify the critical role of proper α parameter selection in the CEBlock's performance across diverse architectural backbones.

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3) Effects of Different Scales: To further validate the effectiveness across different scales, Table VIII reveals consistent performance patterns across all architectures. For CNN-based models, when the scale level is set to 2 (low), the key performance metrics $F1$/IoU decrease by 1.08%/1.71%, whereas setting the scale level to 2 (high) results in reductions of 1.16%/1.83% in $F1$/IoU. On Mamba-based architecture, scale level 2 (low) causes $F1$/IoU to decrease by 2.36%/3.27%, while level 2 (high) leads to reductions of 1.06%/1.47%. For Transformer-based models, scale level 2 (low) results in 0.59%/0.92% decreases in $F1$/IoU, while scale level 2 (high) causes more substantial reductions of 3.74%/5.56%. These experimental results conclusively demonstrate the importance of appropriate multiscale configuration in the CEBlock architecture across all three model types, with multilevel consistently delivering optimal performance.

We also visualized the features extracted at each stage by the disassembled architecture of the CEBlock in Fig. 8. Among the features extracted at each stage, the features in the high-response regions are not necessarily the regions of interest, that is, the changed regions. Compared with the baseline, the more the number of levels in the CEBlock and the closer the α configuration is to 0.1, the more balanced the features extracted at each stage, and the fewer false detections of the regions of interest.

V. Conclusion

This study proposes a plug-and-play block for RSCD, which has been evaluated on benchmark datasets such as WHU, DSIFN, and SYSU. The proposed method exhibits performance advantages in capturing both subtle and large-scale changes in different landscapes. By calculating the custom contrastive learning loss of the bitemporal image features extracted by the Siamese network, the proposed CEBlock enhances the ability of the original methods to extract more discriminative and robust features. The design principles of this architecture provide valuable insights for the development of change detection systems in practical applications, including urban monitoring, disaster assessment, and land-cover mapping. Looking ahead, we believe that contrastive learning will gain increasing attention in remote sensing image processing.

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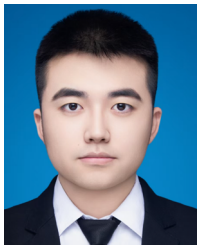
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